Derivation and Implementation Notes

Woloshyn — Demonstrating Quantum Computing with the Quark Model
arXiv:2301.10828v2

Companion document for the `quarksim` codebase

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1 The Physics Problem

1.1 Charmonium with Spin-Dependent Interactions

This paper extends the quantum computing approach to charmonium by including **spin-dependent interactions**, which break the degeneracy between spin-singlet and spin-triplet states. Three channels are studied:

- $^1S_0$ (η_c): orbital $l = 0$, spin singlet ($\vec{S}_c \cdot \vec{S}_{\bar{c}} = -3/4$)
- $^3S_1$ (J/ψ): orbital $l = 0$, spin triplet ($\vec{S}_c \cdot \vec{S}_{\bar{c}} = +1/4$)
- $^1P_1$ (h_c): orbital $l = 1$, spin singlet

1.2 The Cornell Potential with Spin Contact Term

The potential between the charm quark and anti-charm quark is:

$$V(r) = -\frac{a}{r} + b\,r + V_s(r)\,\vec{S}_c \cdot \vec{S}_{\bar{c}} \tag{1}$$

where $a = 4\alpha_s/3$ is the color Coulomb coupling, b is the string tension, and $V_s(r)$ is the spin-dependent contact potential:

$$V_s(r) = \frac{32\pi\alpha_s}{9\,m_c^2}\,\delta(r) \tag{2}$$

The delta function is smeared with a Gaussian:

$$\delta(r) = \left(\frac{\sigma}{\sqrt{\pi}}\right)^3 e^{-\sigma^2 r^2} \tag{3}$$

Note that spin-orbit and tensor interactions are omitted as they are not relevant for the states considered here.

Contrast with Gallimore: The Gallimore implementation uses a spin-averaged potential ($V = -\kappa/r + \sigma r$) and considers only spin-averaged S-wave states. Woloshyn's inclusion of the contact term enables the study of the physical mass splitting between η_c (2982 MeV) and J/ψ (3097 MeV), a difference of 115 MeV.

1.3 Parameters

The parameters from Barnes *et al.* (Table 1 of the paper):

Symbol	Value	Description
α_s	0.5461	Strong coupling constant
b	0.1425 GeV^2	String tension
m_c	1.4794 GeV	Charm quark mass
σ	1.0946 GeV	Gaussian smearing parameter
ω	1.2 fm^{-1}	HO frequency (chosen from Fig. 1)

Units: Unlike Gallimore (which works in MeV), all energies here are in fm^{-1}. The conversion factor is $\hbar c = 197.326$ MeV·fm, so 1 fm^{-1} = 197.3 MeV. The meson mass is $M = E + 2m_c$, where E is the energy eigenvalue.

Code: `hamiltonian.py`, constants in `CHARMONIUM_PARAMS`.

2 Harmonic Oscillator Basis

2.1 Basis States

The Hamiltonian for the 3D harmonic oscillator is (Eq. 2):

$$H_{\mathrm{HO}} = -\frac{\vec{\nabla}^2}{2\mu} + \frac{1}{2}\mu\omega^2 r^2 \tag{4}$$

The quark model Hamiltonian is rewritten as (Eq. 3):

$$H_{\mathrm{qm}} = H_{\mathrm{HO}} + V(r) - \frac{1}{2}\mu\omega^2 r^2 \tag{5}$$

This decomposition is convenient because the matrix elements of H_{HO} are trivially diagonal, and $V(r) - \frac{1}{2}\mu\omega^2 r^2$ contains the "residual" interaction.

The HO basis wavefunctions (unnormalized) are:

$$r^l\, e^{-\frac{1}{2}\nu r^2}\, L_n^{l+1/2}(\nu r^2)\, Y_{lm}(\theta, \phi) \tag{6}$$

where $\nu = \mu\omega$, $L_n^{l+1/2}$ is a generalized Laguerre polynomial, and the eigenenergy is $\omega(2n+l+3/2)$ with $\hbar = 1$.

2.2 Truncation to $N = 4$ States

The basis is truncated to $N = 4$ radial states ($n = 0, 1, 2, 3$) for each channel. This gives a 4×4 Hamiltonian matrix encodable in **2 qubits** — half the qubits needed for Gallimore's 3-orbital system.

The oscillator parameter $\omega = 1.2$ fm^{-1} is chosen as a compromise: small ω gives better convergence for low-lying states (closer to the exact Schrödinger solution) but worse for higher states. The paper's Fig. 1 shows the eigenvalues vs. ω, demonstrating the variational nature of the truncation.

Code: The matrix is built in `hamiltonian.get_matrix(channel)` using the paper's explicit matrices, or computed from scratch via `hamiltonian.compute_matrix(channel)`.

3 Matrix Elements

3.1 Structure

The matrix elements of H_{qm} in the HO basis are:

$$\langle m|H_{\mathrm{qm}}|n\rangle = \omega\big(2n + l + \tfrac{3}{2}\big)\,\delta_{mn} + \langle m|V(r) - \tfrac{1}{2}\mu\omega^2 r^2|n\rangle \tag{7}$$

The first term is the diagonal HO eigenvalue. The second term contains:

- $-a\,\langle m|r^{-1}|n\rangle$: Coulomb integral (hypergeometric functions, same as Gallimore)
- $b\,\langle m|r|n\rangle$: Linear confinement integral (hypergeometric functions)
- $-\frac{1}{2}\mu\omega^2\,\langle m|r^2|n\rangle$: HO potential subtraction (tridiagonal, closed form)
- $V_s \cdot S \cdot S \cdot \langle m|\delta(r)|n\rangle$: Spin contact integral (numerical quadrature, S-waves only)

3.2 Explicit Matrices

The $^1S_0$ Hamiltonian matrix at $\omega = 1.2$ fm^{-1} (Eq. 4):

$$H_{^1S_0} = \begin{pmatrix} 0.9431 & -0.8733 & -0.7690 & -0.5601 \\ -0.8733 & 3.3365 & -0.5646 & -0.8648 \\ -0.7690 & -0.5646 & 5.4382 & -0.1566 \\ -0.5601 & -0.8648 & -0.1566 & 7.3451 \end{pmatrix} \tag{8}$$

The $^3S_1$ matrix (Eq. 15 in the Appendix):

$$H_{^3S_1} = \begin{pmatrix} 1.0946 & -0.7114 & -0.6111 & -0.4112 \\ -0.7114 & 3.5406 & -0.3910 & -0.6989 \\ -0.6111 & -0.3910 & 5.6122 & -0.0119 \\ -0.4112 & -0.6989 & -0.0119 & 7.5104 \end{pmatrix} \tag{9}$$

The difference between $^1S_0$ and $^3S_1$ on the diagonal is due to the spin contact interaction: $\Delta H_{nn} = V_s \cdot [\frac{1}{4} - (-\frac{3}{4})] \cdot \langle n|\delta(r)|n\rangle$. For $n = 0$: $\Delta = 1.0946 - 0.9431 = 0.1515\,\text{fm}^{-1}$, corresponding to 30 MeV of hyperfine splitting per level.

3.3 Eigenvalues (Table 3)

State	$^1S_0$ (fm^{-1})	$^3S_1$ (fm^{-1})	$^1P_1$ (fm^{-1})
1	0.395	0.753	2.783
2	3.506	3.634	4.875
3	5.664	5.723	6.765
4	7.546	7.648	8.767

These are the target values for the VQITE algorithm.

Code: Eigenvalues via `numpy.linalg.eigvalsh(get_matrix(channel))`.

4 Hamiltonian in Pauli Form (2 Qubits)

4.1 Direct Encoding

With 4 basis states encoded in 2 qubits, the mapping is direct:

$$|00\rangle \leftrightarrow \text{HO state } n = 0$$
$$|01\rangle \leftrightarrow \text{HO state } n = 1$$
$$|10\rangle \leftrightarrow \text{HO state } n = 2$$
$$|11\rangle \leftrightarrow \text{HO state } n = 3$$

Key difference from Gallimore: In the Jordan-Wigner encoding (3 qubits), only 3 of the $2^3 = 8$ computational basis states are physical (the 1-particle sector). Here, *all* $2^2 = 4$ states are physical. This means:

- No need to project onto a physical subspace.
- The full 4×4 Pauli matrix equals the physical Hamiltonian directly.
- Fewer qubits (2 vs. 3) for more basis states (4 vs. 3).

4.2 Pauli Decomposition

A general 4×4 real symmetric matrix decomposes into 10 Pauli terms (the 6 imaginary operators XY, YX, ZY, YZ, IY, YI vanish for real matrices):

$$H = c_0\, II + c_1\, Z_0 + c_2\, Z_1 + c_3\, Z_0Z_1 + c_4\, X_0 + c_5\, X_1 + c_6\, Z_0X_1 + c_7\, X_0Z_1 + c_8\, X_0X_1 + c_9\, Y_0Y_1 \tag{10}$$

The coefficients are computed as:

$$c_k = \frac{1}{4}\text{Tr}(H \cdot P_k) \tag{11}$$

4.3 Pauli Coefficients

Term	$^1S_0$	$^3S_1$	$^1P_1$
c_0 (II)	4.273	4.439	5.798
c_1 (Z_0)	−2.119	−2.122	−1.910
c_2 (Z_1)	−1.082	−1.086	−0.964
c_3 (Z_0Z_1)	−0.129	−0.137	−0.067
c_4 (X_0)	−0.817	−0.655	−0.446
c_5 (X_1)	−0.515	−0.350	0.083
c_6 (Z_0X_1)	−0.358	−0.361	−0.323
c_7 (X_0Z_1)	0.048	0.044	−0.064
c_8 (X_0X_1)	−0.562	−0.401	−0.095
c_9 (Y_0Y_1)	−0.002	0.010	0.133

Note on qubit ordering: The paper uses PennyLane (big-endian: qubit 0 = MSB), while our Qiskit implementation uses little-endian (qubit 0 = LSB). The paper's Z_0 corresponds to Qiskit's `'ZI'`, and Z_1 to `'IZ'`. The coefficients are physically identical; only the labeling convention differs.

Code: `hamiltonian.build_pauli_hamiltonian(channel)` decomposes via `matrix_to_pauli(H)`. Validation via `validate_pauli_hamiltonian()` confirms the 4×4 Pauli matrix reconstructs the original matrix exactly.

5 Variational Quantum Imaginary Time Evolution

5.1 Imaginary Time Evolution

For a Hamiltonian H with eigenvalues $\{\lambda_i\}$ and eigenstates $\{|\phi_i\rangle\}$, the imaginary time evolution of an initial state $|\psi_0\rangle$ is (Eq. 5):

$$e^{-H\tau}\,|\psi_0\rangle = \sum_i e^{-\lambda_i \tau}\,\langle\phi_i|\psi_0|\phi_i|\psi_0\rangle\,|\phi_i\rangle \tag{12}$$

For large τ, the lowest-energy component dominates. However, this operation is **non-unitary** and cannot be directly implemented on a quantum computer.

5.2 The Variational Approach

We parametrize the state as $\left|\psi(\vec{\theta})\right\rangle = U(\vec{\theta})\,|0\rangle$ and seek unitary updates that approximate imaginary time evolution. The variational principle (Eqs. 7–8) leads to the evolution equation (Eq. 9):

$$\boxed{\sum_j A_{ij}\,\dot{\theta}_j = C_i} \tag{13}$$

where:

$$C_i = -\frac{\partial E_\tau}{\partial \theta_i} \tag{14}$$

$$A_{ij} = -\frac{\partial^2 \left\langle \psi(\tilde{\theta})|\psi(\vec{\theta})\middle|\psi(\tilde{\theta})|\psi(\vec{\theta})\right\rangle}{\partial\theta_i\,\partial\theta_j} \tag{15}$$

Here $E_\tau = \left\langle \psi(\vec{\theta}(\tau))\middle|H\middle|\psi(\vec{\theta}(\tau))\right\rangle$ is the energy, and the derivative in (15) acts only on the ket (right side). The matrix A is the **quantum metric tensor** (Fubini-Study metric), which captures the geometry of the parameter space.

Parameters are updated as:

$$\vec{\theta}(\tau + \Delta\tau) = \vec{\theta}(\tau) + \Delta\tau\,\dot{\vec{\theta}} \tag{16}$$

where $\Delta\tau$ acts as a learning rate.

5.3 The Ansatz Circuit (Fig. 2)

The variational ansatz uses 3 parameters on 2 qubits:

```
q0:  -Ry(θ0)-*------
                |
q1:  -Ry(θ1)-X-Ry(θ2)-
```

With 3 parameters, this circuit can produce any real superposition of 4 basis states (3 free parameters = 4 real amplitudes − 1 normalization − 0 phases since all are real).

Starting from $|00\rangle$:

$$\begin{aligned}|\psi(\theta_0,\theta_1,\theta_2)\rangle = \cos\tfrac{\theta_0}{2}\left(\cos\tfrac{\theta_1+\theta_2}{2}\,|00\rangle + \sin\tfrac{\theta_1+\theta_2}{2}\,|10\rangle\right)\\ + \sin\tfrac{\theta_0}{2}\left(\sin\tfrac{\theta_1-\theta_2}{2}\,|01\rangle + \cos\tfrac{\theta_1-\theta_2}{2}\,|11\rangle\right)\end{aligned} \tag{17}$$

(Note: state indices use Qiskit's $|q_1\,q_0\rangle$ ordering.)

5.4 Computing C_i and A_{ij}

Energy gradient C_i: We use the **parameter-shift rule** for RY gates:

$$\frac{\partial E}{\partial \theta_k} = \frac{E(\theta_k + \pi/2) - E(\theta_k - \pi/2)}{2} \tag{18}$$

This is exact (not a finite-difference approximation) because $RY(\theta) = e^{-i\theta Y/2}$ generates a sinusoidal parameter dependence.

Metric tensor A_{ij}: We use **finite differences** on the overlap function $f(\vec{\theta}) = \left\langle \psi_0|\psi(\vec{\theta})\middle|\psi_0|\psi(\vec{\theta})\right\rangle$:

$$A_{ij} \approx -\frac{f(\vec{\theta}+\varepsilon\hat{e}_i+\varepsilon\hat{e}_j) - f(\vec{\theta}+\varepsilon\hat{e}_i-\varepsilon\hat{e}_j) - f(\vec{\theta}-\varepsilon\hat{e}_i+\varepsilon\hat{e}_j) + f(\vec{\theta}-\varepsilon\hat{e}_i-\varepsilon\hat{e}_j)}{4\varepsilon^2} \tag{19}$$

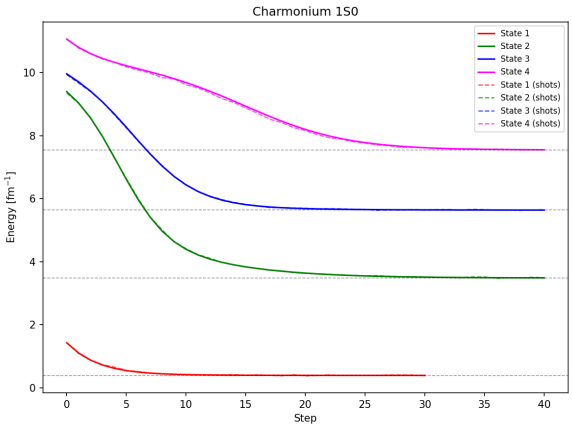
with $\varepsilon = 10^{-4}$. The matrix A is regularized as $A + \lambda I$ ($\lambda = 10^{-4}$) to handle near-singular cases when the state is close to a critical point in parameter space.

5.5 Three Simulation Modes

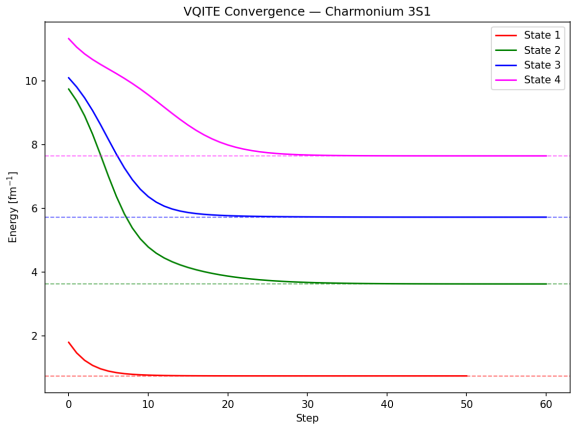
Our implementation supports three simulation modes, matching the paper:

1. **Wave function simulator** (`shots=0`): Exact statevector evaluation via `Statevector.expectation_valu` No circuits are executed. This is equivalent to the paper's PennyLane `default.qubit` with `shots=None`.
2. **Ideal quantum simulation** (`shots>0`, `noise=0`): Each Pauli term is measured by running an actual circuit on the Qiskit `AerSimulator` with finite shots. This introduces **shot noise** (statistical sampling uncertainty) but no systematic errors.
3. **Noisy simulation** (`shots>0`, `noise>0`): Same as mode 2 but with a **depolarizing noise model** applied to every gate. This simulates the systematic errors present on real quantum hardware.

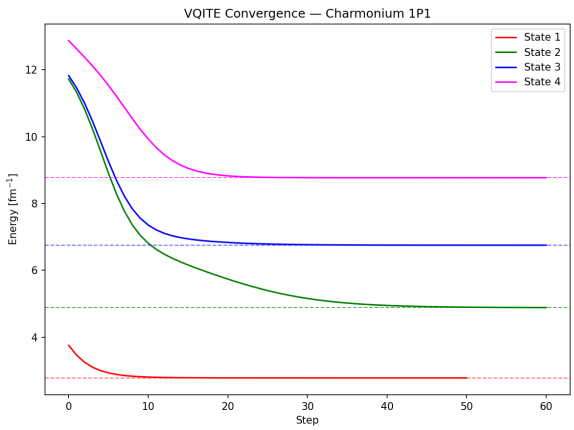
In all modes, the energy gradient C uses the corresponding energy estimator (exact or shot-based). The metric tensor A is always computed exactly via statevector—this is cheap for 2 qubits and avoids the numerical instability caused by noisy matrix inversion.



(a) $^1S_0$ channel



(b) $^3S_1$ channel



(c) $^1P_1$ channel

Figure 1: VQITE convergence for all three channels (Figs. 4–6 of the paper). **Solid lines**: wave function simulator (exact statevector). **Colored dashed lines**: ideal quantum simulation with 20,000 shots per Pauli term. **Horizontal gray lines**: eigenvalues from diagonalizing the truncated Hamiltonian. Parameters: $\Delta\tau = 0.02$, initial $\theta_i = 0.5$, 50 steps.

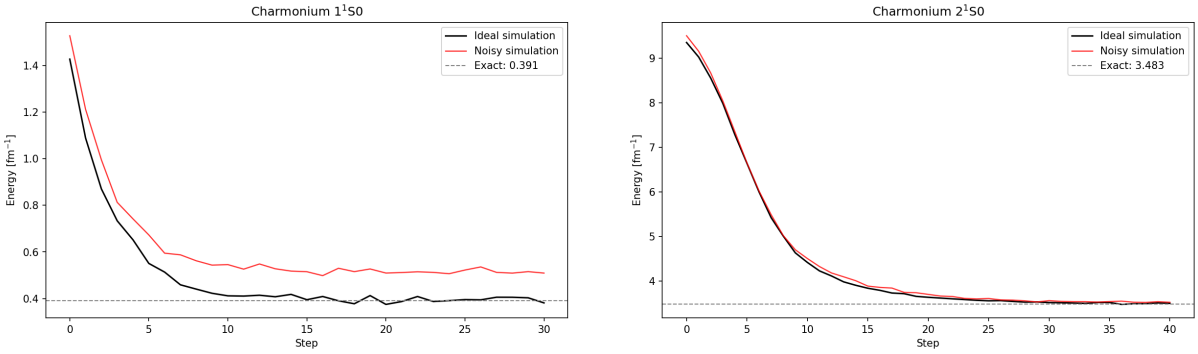
5.6 Convergence

The ground state converges within ∼20–30 steps. The shot-based curves closely track the exact ones with visible fluctuations from sampling noise. Excited states (computed via the penalty method, Section 6) take slightly longer.

5.7 Effect of Gate Noise

Under depolarizing noise, the measured energy is biased upward. For a depolarizing rate p per gate, each layer of the circuit has a success probability $r = 1-p$, and the noisy expectation value of a traceless Pauli operator scales as $\langle P\rangle_{\text{noisy}} \approx r^d \langle P\rangle_{\text{ideal}}$, where d is the circuit depth. This shrinks the signal toward zero, which for the Hamiltonian means the energy is pushed toward the trace (average eigenvalue)—always *above* the ground state.

Code: `vqite.run_vqite()` for exact mode. `vqite.run_vqite_shot_based()` for shot-based and noisy modes, using `simulation._estimate_energy_shot_based()` and `simulation.make_noise_model()` from the shared infrastructure.



(a) 1^1S_0 ground state

(b) 2^1S_0 first excited state

Figure 2: Effect of depolarizing gate noise on VQITE convergence for the $^1S_0$ channel (compare with Figs. 9–10 of the paper). **Black**: ideal shot-based simulation (20,000 shots, no gate errors). **Red**: noisy simulation with 1% depolarizing error per gate. **Dashed gray**: exact eigenvalue. The noisy simulation converges to a systematically higher energy because the depolarizing channel pushes the state toward the maximally mixed state.

6 Excited States

6.1 Penalty Method (Eq. 12)

To find the k-th excited state, we add a penalty term to the energy that pushes the variational state away from all lower-lying states:

$$E_{\text{eff}} = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle + \alpha \sum_{j<k} \left| \left\langle \phi_j | \psi(\vec{\theta}) \middle| \phi_j | \psi(\vec{\theta}) \right\rangle \right|^2 \tag{20}$$

where $|\phi_j\rangle$ are the previously determined eigenstates and α is a large positive coefficient.

Contrast with Gallimore: Gallimore uses an explicit orthogonalization in parameter space (reducing from 2 to 1 parameter for the first excited state). Woloshyn's penalty method is more general: it works for any number of excited states and any ansatz, at the cost of choosing α large enough.

In practice, $\alpha = 10$ works well for all channels. The overlap terms can be computed via the $U_i U_f^\dagger$ circuit (Fig. 3 of the paper) or directly from statevectors.

6.2 Sequential Computation

States are computed in order:

1. Ground state: VQITE with H
2. 1st excited: VQITE with $H + \alpha |\langle \phi_0 | \cdot | \phi_0 | \cdot \rangle|^2$
3. 2nd excited: VQITE with $H + \alpha \left(|\langle \phi_0 | \cdot | \phi_0 | \cdot \rangle|^2 + |\langle \phi_1 | \cdot | \phi_1 | \cdot \rangle|^2 \right)$
4. 3rd excited: determined by the remaining orthogonal direction (or VQITE with 3 penalty terms)

Code: `vqite.run_vqite_excited()` takes a list of lower-state wavefunctions and adds the penalty.

7 M1 Transition Amplitudes

7.1 Physics

M1 transitions connect spin-singlet and spin-triplet states with the same orbital angular momentum (e.g., $^3S_1 \to {}^1S_0$). In the nonrelativistic limit, the transition involves a spin flip, and

the spatial part of the amplitude is simply the wavefunction overlap:

$$\mathcal{A}_{M1} = \langle \psi_f | \psi_i | \psi_f | \psi_i \rangle \,, \qquad |\mathcal{A}_{M1}|^2 = \text{squared amplitude} \tag{21}$$

Because the spin-dependent interaction slightly distorts the spatial wavefunctions in different channels, the overlap is close to (but not exactly) 1 for corresponding states and close to 0 for non-corresponding states.

7.2 Quantum Circuits

Overlap circuit (Fig. 3): Prepare $|\psi_i\rangle$ then apply $U_f^\dagger$. Measuring $|00\rangle$ gives $P(00) = |\langle \psi_f | \psi_i | \psi_f | \psi_i \rangle|^2$.

Swap test (Fig. 7): Uses an ancilla qubit and controlled-SWAP. The measurement yields:

$$P(\text{ancilla} = 0) = \frac{1 + |\langle \psi_f | \psi_i | \psi_f | \psi_i \rangle|^2}{2} \tag{22}$$

This is less precise for small overlaps but doesn't require implementing $U^\dagger$.

7.3 Results

Transition	Paper (exact)	Our w.f. sim
$1^3S_1 \to 1^1S_0$	0.9826	0.994
$2^3S_1 \to 1^1S_0$	0.0107	0.004
$2^3S_1 \to 2^1S_0$	0.9781	0.994
$3^3S_1 \to 2^1S_0$	0.0061	0.002

The "exact" column uses solutions of the Schrödinger equation (Table 2 of the paper), while our results use the truncated basis. The dominant overlaps (~ 1) match well; the small off-diagonal overlaps differ because they are sensitive to basis truncation.

Code: `transitions.m1_overlap()` for statevector computation. `transitions.build_overlap_circuit()` and `transitions.build_swap_test_circuit()` for the quantum circuit versions.

8 E1 Transition Amplitudes

8.1 Physics

E1 transitions connect states with $\Delta L = 1$ (S-wave $\leftrightarrow$ P-wave) and the same spin. The transition amplitude is governed by the radial matrix element:

$$\langle u_f(r) | r | u_i(r) | u_f(r) | r | u_i(r) \rangle = \int_0^\infty u_f(r)\, r\, u_i(r)\, dr \tag{23}$$

where u_f is a P-wave ($l = 1$) state and u_i is an S-wave ($l = 0$) state.

8.2 Matrix Elements of r (Eq. 13)

The matrix $\langle m_P | r | n_S \rangle$ connecting P-wave and S-wave HO basis states is (in fm):

$$R = \begin{pmatrix} 0.578 & -0.472 & 0 & 0 \\ 0 & 0.746 & -0.667 & 0 \\ 0 & 0 & 0.882 & -0.817 \\ 0 & 0 & 0 & 1.000 \end{pmatrix} \tag{24}$$

The tridiagonal structure reflects the selection rule: HO states only couple to adjacent radial quantum numbers via r.

8.3 Pauli Decomposition (Eq. 14)

The r-operator matrix has **complex Pauli coefficients** (unlike the Hamiltonian which is real symmetric). The decomposition includes Y terms:

$$\hat{r} = c_0 + c_1 Z_0 + c_2 Z_1 + c_3 Z_0 Z_1 + c_4 (X_0 X_1 + Y_0 Y_1) + c_5 (X_1 - iY_1) + c_6 Z_0 (X_1 + iY_1) + c_7 (X_0 Y_1 - Y_0 X_1) \tag{25}$$

8.4 Hadamard Test (Fig. 8)

The Hadamard test circuit computes $\left\langle \psi_f | \hat{O} | \psi_i \middle| \psi_f | \hat{O} | \psi_i \right\rangle$ for an arbitrary operator $\hat{O}$. It uses an ancilla qubit, controlled state preparations, and a rotation R to select the real or imaginary part:

- $R = R_y(-\pi/2)$: ancilla measurement gives $\langle \sigma_x | \sigma_x \rangle = 2\,\mathrm{Re} \left\langle \psi_f | \hat{O} | \psi_i \middle| \psi_f | \hat{O} | \psi_i \right\rangle$
- $R = R_x(\pi/2)$: ancilla measurement gives $\langle \sigma_y | \sigma_y \rangle = 2\,\mathrm{Im} \left\langle \psi_f | \hat{O} | \psi_i \middle| \psi_f | \hat{O} | \psi_i \right\rangle$

Code: `transitions.e1_matrix_elements()` computes R. `transitions.e1_amplitude()` does statevector computation. `transitions.build_hadamard_test_circuit()` builds the circuit.

9 Readout Error Mitigation

9.1 The Problem

On real quantum hardware, qubits may be misidentified during measurement: $|0\rangle$ read as $|1\rangle$ and vice versa. For n qubits, the readout error is characterized by a $2^n \times 2^n$ calibration matrix M:

$$\vec{p}_{\text{noisy}} = M \cdot \vec{p}_{\text{ideal}} \tag{26}$$

where $\vec{p}$ is the vector of measurement probabilities.

9.2 Mitigation Strategy

The mitigation procedure (described in the paper's Section 5):

1. **Calibrate:** Prepare each computational basis state $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ and measure. The resulting statistics form the columns of M.
2. **Correct:** Apply M^{-1} to the measured probability vector to recover the ideal distribution.

For 2 qubits, M is only 4×4, making inversion straightforward. The paper uses the IBM `ibmq_manila` noise model for demonstration.

9.3 Effect on Results

The readout error mitigation is most impactful for:

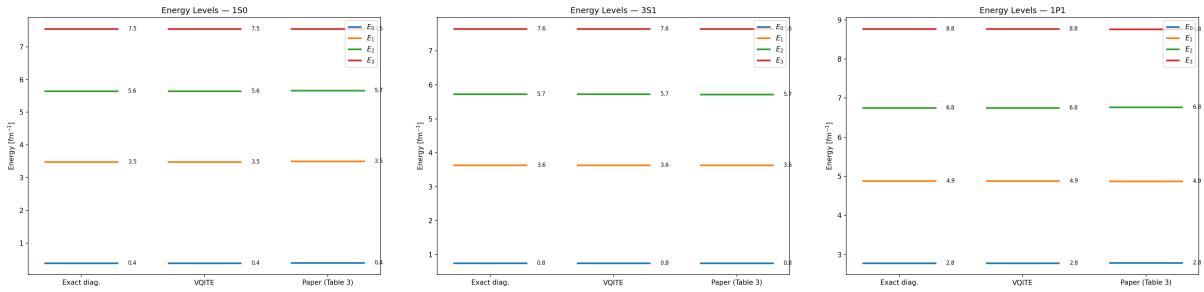
- **Energy calculations**: Figs. 9–11 show ideal vs. noisy vs. mitigated VQITE convergence. Mitigation brings noisy results close to ideal.
- **M1 overlaps**: Table 6 shows that large overlaps (~ 1) are well-recovered, but small overlaps (~ 0.001) remain unreliable even after mitigation.

10 Results Summary

10.1 Energy Levels

Our VQITE simulation reproduces the paper's eigenvalues (Table 3) to within the truncation error:

	$^1S_0$		$^3S_1$	
State	VQITE	Paper	VQITE	Paper
1	0.391	0.395	0.752	0.753
2	3.483	3.506	3.633	3.634
3	5.643	5.664	5.727	5.723
4	7.545	7.546	7.646	7.648



(a) $^1S_0$ (b) $^3S_1$ (c) $^1P_1$

Figure 3: Energy level comparison: VQITE results vs. exact diagonalization vs. paper's Table 3 values.

11 Code–Equation Mapping

Paper reference	Code location	Description
Eq. (1): Schrödinger eq.	`hamiltonian.compute_matrix()`	From-scratch computation
Eq. (3): H_{qm}	`hamiltonian.get_matrix()`	Paper's explicit matrices
Eq. (4): $^1S_0$ matrix	`PAPER_MATRICES["1S0"]`	Hardcoded constant
Section 2.2.1: Pauli form	`hamiltonian.matrix_to_pauli()`	$c_k = \text{Tr}(HP_k)/4$
Fig. 2: Ansatz circuit	`ansatz.build_ansatz()`	3-param Ry+CNOT
Eq. (9): VQITE update	`vqite.run_vqite()`	Full evolution loop
Eq. (10): Gradient C_i	`vqite.compute_energy_gradient()`	Parameter-shift rule
Eq. (11): Metric A_{ij}	`vqite.compute_metric_tensor()`	Finite differences
Eq. (12): Excited penalty	`vqite.run_vqite_excited()`	$E + \alpha\lvert\langle\phi\lvert\psi\rvert\phi\lvert\psi\rangle\rvert^2$
Eq. (13): $\langle m_P\lvert r\rvert n_S\rangle$	`transitions.e1_matrix_elements()`	Numerical integration
Eq. (14): r in Pauli form	`transitions.build_e1_pauli_operator()`	Complex coefficients
Fig. 3: Overlap circuit	`transitions.build_overlap_circuit()`	$U_i U_f^\dagger$
Fig. 7: Swap test	`transitions.build_swap_test_circuit()`	Ancilla + CSWAP
Fig. 8: Hadamard test	`transitions.build_hadamard_test_circuit()`	$\langle\psi_f\lvert O\rvert\psi_i\rvert\psi_f\lvert O\rvert\psi_i\rangle$

12 Running the Simulation

```
# Install
uv venv && source .venv/bin/activate
uv pip install -e .

# Run all channels
python -m quarksim.woloshyn.run --channel all --transitions
```

```
# Single channel, custom parameters
python -m quarksim.woloshyn.run --channel 1S0 --n-steps 80 --dtau 0.01

# Output directory
python -m quarksim.woloshyn.run --output results/woloshyn
```

Output files:

- `vqite_result_{channel}.json` — full simulation record per channel
- `vqite_convergence_{channel}.png` — energy vs. step (Fig. 1)
- `energy_levels_{channel}.png` — level comparison (Fig. 3)
- `wavefunction_{channel}.png` — ground state amplitudes